

Dr. Doc:

Welcome to pharmacology class 🎓

🫁💊 DRUGS FOR DISORDERS OF THE RESPIRATORY SYSTEM

🫁 Asthma Overview

📖 Definition:

▼ A chronic inflammatory disease.

🔬 Pathophysiology:

✅ Characterized by

✂️ Airway inflammation,

🌀 Inflammation causes: Mucosal edema, Increased mucus production

✂️ Bronchial hyper-reactivity, and

✂️ Bronchospasm.

🧠 Major Symptoms of Asthma

🫁 Symptoms include

 shortness of breath,

 cough,

 chest tightness,

 wheezing, and

 rapid respiration.

 Classification of Asthma Severity

- ◆ Asthma is classified as Intermittent or Persistent (Mild, Moderate, Severe).

 Criteria for Severity:

✓Includes frequency of symptoms,

✓nighttime awakenings,

✓frequency of SABA use for relief, interference with normal activity, and lung function tests (FEV1 and FEV1/FVC ratio).

 Treatment Goals and Drug Classes

 Goals of Therapy:

 Decrease intensity/frequency of symptoms,

 prevent future exacerbations, and

 minimize activity limitations.

 Drug Categories:

 Anti-inflammatory drugs and

 Bronchodilators.

- 🚫 Administration:

^ Most treatments are administered via inhalation.

📌 Classification of Bronchial Asthma Drugs

- 📌 Bronchodilators: Include

● β_2 Sympathomimetics

● Methylxanthines, and

● Anticholinergics.

- 📌 Anti-inflammatory Agents: Include

● Leukotriene(LT#) antagonists,

● Mast cell stabilizers,

● Corticosteroids (Systemic and Inhalational), and

● Anti-IgE antibodies.

▲ Bronchodilators Mechanism of Action 📌

+ Stimulate β_2 -receptors

📌 β_2 receptors, increasing cAMP formation in bronchial muscle cells, which leads to muscle relaxation.

⚠ Note: These drugs provide symptomatic relief but have no anti-inflammatory effects.

○ Short-acting β_2 Agonists (SABAs)

📌 Use:

✅ For quick relief of symptoms.

🕒 Pharmacokinetics:

⚡ Rapid onset (5 to 30 minutes) with relief lasting 4 to 6 hours.

🫁 Indications:

✅ Appropriate for mild, intermittent asthma or exercise-induced bronchospasm (EIB).

❌ They should not be used as monotherapy for persistent asthma.

📌 Examples: Albuterol (Salbutamol), Levalbuterol, and Terbutaline.

▼ Long-acting β_2 Agonists (LABAs)

🕒 Action:

✅ Provide bronchodilation for at least 12 hours and are taken on a fixed schedule.

⚠️ Requirement:

✅ Must only be used in combination with an asthma controller, such as an inhaled corticosteroid (ICS).

📌 Examples: Salmeterol and Formoterol

⌚ Ultra-long-acting:

✅ Indacaterol, Olodaterol, Vilanterol, and Bambuterol (primarily for COPD).

❌ Adverse Effects of β_2 -agonists

💓 Side effects include tachycardia/angina

 hyperglycemia,

 hypokalemia, and

 hypomagnesemia.

 Mitigation:

 These adverse effects are minimized when the medication is inhaled.

 Anti-cholinergic Drugs

 Use: Primarily used for COPD.

 Examples:

 Ipratropium (short-acting, 4-6 hours) and

 Tiotropium (long-acting, 24 hours).

 Mechanism:

 Block muscarinic cholinergic receptors in the bronchi to prevent bronchoconstriction and mucus secretion.

 Adverse Effects:

 Xerostomia (dry mouth) and a bitter taste.

 Methylxanthines (Theophylline) Mechanism

 Mechanism of Action:

🚫 Inhibition of phosphodiesterase (PDE), which increases cyclic nucleotide concentrations.

🚫 Blockade of adenosine receptors.

💉 Release of Ca^{2+} from the sarcoplasmic reticulum.

🔗 Result: Bronchodilatation in the bronchial muscle.

💊 Theophylline Pharmacokinetics

💧 Well absorbed orally and 50% protein-bound.

🧬 Metabolism: Extensively metabolized in the liver (demethylation and oxidation) by CYP1A2.

🕒 Half-life ($t_{1/2}$):

7–12 hours in adults, 3–5 hours in children, and slower in the elderly.

📄 Theophylline Safety and Interactions

🚨 Safety:

Has a narrow margin of safety; side effects include headache, nervousness, nausea, cardiac arrhythmias, and seizures.

💊 Drug Interactions:

 Metabolism Enhancers:

Phenytoin, rifampicin, and phenobarbitone.

 Metabolism Inhibitors :

Erythromycin, ciprofloxacin, and cimetidine.

 Glucocorticoids Introduction and MoA

 Role:

Drugs of choice for long-term control of airway inflammation.

 Mechanism:

 Inhibit Phospholipase A2 to prevent the release of arachidonic acid, leading to direct anti-inflammatory effects.

 Additional Effects:

 Decrease the inflammatory cascade (eosinophils, macrophages, and leukocytes).

 Reduce airway mucosal edema and mucus production.

 Increase the number and responsiveness of bronchial β_2 receptors.

 Glucocorticoid Formulations and Status Asthmaticus

 Inhaled Glucocorticoids (IGC):

▼ Examples include

✓ beclomethasone, budesonide, fluticasone, mometasone, and triamcinolone.

💊 Oral Corticosteroids (CS):

Examples include methylprednisolone, prednisone, and prednisolone.

🚒 Status Asthmaticus:

Severe exacerbations may require intravenous methylprednisolone or oral prednisone to reduce airway inflammation.

🚫 Adverse Effects of Glucocorticoids

👄 Common Effects:

🎯 Oropharyngeal candidiasis, 🎯 dysphonia, and

🎯 sore throat.

🦴 Systemic Effects (Long-term/High dose):

🚫 Adrenal suppression,

🚫 slow growth in children/adolescents, and

🚫 bone loss.

🔴 Oral-Specific Effects:

⚠️ Hyperglycemia and peptic ulcer disease.

👶 Pregnancy:

☀️ Inhaled steroids are considered safe during pregnancy.

💊 Mast Cell Stabilizers (Sodium Cromoglycate)

🛡️ Action:

🔑 Acts as a prophylactic anti-inflammatory agent by 🚫 inhibiting mast cell degranulation and the subsequent release of histamine, leukotrienes (LTs), and interleukins (ILs).

⚠️ Clarification:

☀️ It is not a bronchodilator and does not antagonize the constrictor actions of histamine, acetylcholine (Ach), or LTs.

🔴 Mast Cell Stabilizer Uses, PK, and AEs

🎯 Therapeutic Uses:

⚔ Used as an alternative therapy, for exercise-induced bronchospasm (EIB), and allergic rhinitis.

💊 Pharmacokinetics (PK):

🚫 It is not absorbed orally,

○ available as a nebulized solution,

○ has a short duration of action, and is rapidly excreted unchanged in urine and bile.

⚠ Adverse Effects (AEs):

Includes cough, throat irritation, and an unpleasant taste.

💊 Leukotriene Modifiers Overview

🧬 Zileuton:

🚫 An inhibitor of 5-lipoxygenase (5-LOX) that prevents the formation of both LTB₄ and cysteinyl leukotrienes.

🚫 Zafirlukast and Montelukast:

Selective antagonists of the cysteinyl leukotriene-1 (cysLT₁) receptor.

🎯 Usage:

Ⓜ They are alternatives to inhaled glucocorticoids for mild-to-moderate asthma and can have an additive effect in severe asthma.

They are effective for aspirin-induced and exercise-induced asthma.

Leukotriene Synthesis Pathway

Pharmacokinetics of Leukotriene Modifiers

 Metabolism:

They are orally active, highly protein-bound, and undergo extensive hepatic metabolism.

- ⌚ Half-life ($t_{1/2}$): Montelukast (3–6 hours), Zafirlukast (8–12 hours), and Zileuton (2.5 hours).

 Excretion:

Zileuton is excreted in urine; Zafirlukast and montelukast undergo biliary excretion.

 Interaction: Food reduces the absorption of zafirlukast.

Adverse Effects and Interactions of Leukotriene Modifiers

 Neuropsychiatric Effects:

These include depression, anxiety, abnormal dreams, hallucinations, insomnia, and suicidal thinking.

Specific Drug Effects:

Zafirlukast can cause headache and GI disturbance. Zileuton is hepatotoxic and inhibits CYP1A2.

Enzyme Inhibition:

Zafirlukast inhibits CYP2C8, 2C9, and 3A4.

Monoclonal Antibodies

Omalizumab:

☒ Selectively binds to human IgE, decreasing its binding to receptors on mast cells and basophils; it is reserved for resistant, allergic asthma.

IL-5 Antagonists:

Mepolizumab, benralizumab, and reslizumab are interleukin-5 antagonists.

Monoclonal Antibodies Indications and Limitations

Indication:

Treatment of severe persistent asthma poorly controlled by conventional therapy.

 Limitations:

 High cost, specific administration routes (IV for reslizumab, SC for others), and a significant adverse effect profile.

 Adverse Effects of Monoclonal Antibodies

 Side effects include

rare anaphylactic reactions, arthralgias, fever, rash, and an increased risk of infections.

Pharmacotherapy of Cough

 Definition:

 Cough is a protective reflex intended to expel secretions and foreign particles.

 Classification:

Can be productive or non-productive; identifying the cause is essential before treatment.

Expectorants and Mucolytics

 Expectorants (e.g., Guaifenesin):

Increase bronchial secretions or reduce viscosity to facilitate removal by coughing.

 Mucolytics (e.g., Hypertonic saline, Acetylcysteine):

React directly with mucus to make it less viscous.

Antitussives and Codeine

 General:

 Antitussives suppress cough centrally and peripherally and should only be used for dry, non-productive coughs.

 Codeine:

 Decreases the sensitivity of the CNS cough center.

↔ At high doses, it can cause respiratory depression and drowsiness (especially in children); AEs include constipation, fatigue, and addiction.

 Pholcodine

 Properties: Similar antitussive efficacy to codeine but has practically no analgesic or addicting properties.

 Duration: It is longer-acting, with effects lasting up to 12 hours.

▼ Opioid Antitussives - Pholcodine

Properties:

✓ It has practically no analgesic or addicting properties.

Efficacy:

Its antitussive efficacy is similar to that of codeine.

Duration:

It is longer-acting, with effects lasting up to 12 hours.

▼ Nonopioid Antitussives - Dextromethorphan

Mechanism:

It is a synthetic central N-methyl D-aspartate (NMDA) receptor antagonist.

Key Characteristics:

✓ It does not depress the mucociliary function of the airway mucosa and is practically free of constipating effects.

Adverse Effects:

🤪 Common effects include dizziness, nausea, and drowsiness; high doses may result in hallucinations and ataxia.

🧠 Adjuvant Antitussives

📌 Antihistamines:

✅ Often added to antitussive or expectorant formulations.

💧 First-generation antihistamines are commonly used, including Chlorpheniramine, Diphenhydramine, and Promethazine.

👉 Bronchodilators: Examples include Salbutamol and Terbutaline.

▼ Drugs Used to Treat Allergic Rhinitis

📖 Definition: Allergic rhinitis is an inflammatory disorder affecting the upper airway.

🤧 Symptoms:

Includes sneezing, rhinorrhea (runny nose), pruritus (itching), and nasal congestion (caused by the dilation and increased permeability of nasal blood vessels).

🌧️ Forms: It exists in two major forms: seasonal and perennial.

📌 Antihistamines for Allergic Rhinitis

 **Treatment Overview:** Options include antihistamines, corticosteroids, and α -adrenergic agonists.

 **H1 Receptor Antagonists:** Second-generation antihistamines are primarily used, such as fexofenadine, loratadine, desloratadine, and cetirizine.

 **Topical Intranasal Antihistamines:** Includes olopatadine and azelastine, which provide increased drug delivery with fewer adverse effects.

 **Corticosteroids for Allergic Rhinitis**

 **Intranasal Formulations:** Examples include beclomethasone, budesonide, fluticasone, ciclesonide, mometasone, and triamcinolone.

 **Onset of Action:**

Ranges from 3 to 36 hours; however, for chronic rhinitis, improvement may not be seen for 1 to 2 weeks after starting therapy.

 **Adverse Effects:**

Nasal irritation, nose bleeds, sore throat, and rarely, candidiasis.

 **α -Adrenergic Agonists (Nasal Decongestants)**

 **Mechanism:**

💧 They activate α 1-adrenoceptors in the nasal mucosa arterioles, causing vasoconstriction, which reduces edema and airway resistance.

📌 Examples: Phenylephrine, pseudoephedrine, and oxymetazoline.

⚠️ Usage Limitations:

Intranasal formulations should not be used for more than 3 days due to the risk of rebound nasal congestion.

📌 Oral vs. Intranasal:

🌐 Oral formulations provide a longer duration of action but also cause increased systemic effects, such as elevated blood pressure, increased heart rate, excitation, and the potential for abuse (specifically pseudoephedrine).

🌈 Theophylline is a methylxanthine derivative (chemically related to caffeine) used primarily as a bronchodilator in chronic asthma and COPD.

💡 Its mechanism of action is multifactorial, involving three distinct cellular targets:

🧠 The Core Mechanisms of Action (The "Big Three")

Mechanism 1: Non-Selective 🟠 Phosphodiesterase (PDE) 🚫 Inhibition (Primary Bronchodilator Pathway)

🎯 Target:

🚫 Inhibits PDE3 and PDE4 enzymes.

~ Biochemical Cascade:

1. PDE normally degrades intracellular cyclic AMP (cAMP) into 5'-AMP.
2. By blocking PDE, theophylline causes an accumulation of cAMP inside bronchial smooth muscle cells and inflammatory cells.
3. Increased cAMP activates Protein Kinase A (PKA).
4. PKA phosphorylates and  inactivates Myosin Light Chain Kinase (MLCK) while promoting calcium efflux out of the cytoplasm.

Result:

 Bronchial smooth muscle relaxation (bronchodilation) and decreased release of inflammatory mediators from mast cells and eosinophils.

Mechanism 2: Competitive Adenosine Receptor Antagonism (The "Caffeine-Like" Pathway)

Target:

- ▼ Blocks  cell-surface Adenosine (A₁ and A₂) receptors.

~ Physiological Effect:

 In asthmatic patients, endogenous adenosine causes smooth muscle bronchoconstriction and triggers mast cells to release histamine and leukotrienes.

➡ By blocking adenosine receptors, theophylline prevents adenosine-induced bronchospasm.

✗ Side Effect Link

* Adenosine normally acts as an inhibitory neurotransmitter in the CNS and slows cardiac conduction.

➡ Blocking adenosine receptors explains theophylline's classic toxic side effects:

✗ CNS 🧠 stimulation (seizures/insomnia) and

✗ cardiac 🫀 stimulation (tachycardia/arrhythmias).

Mechanism 3:

Activation of Histone Deacetylase 2 (HDAC2) (Anti-Inflammatory Pathway)

🎯 Target:

😊 Recruits and + activates HDAC2 in inflammatory cell nuclei.

○ Biochemical Cascade:

1. Inflammatory cytokines cause acetylation of histones, keeping DNA open for transcription of more inflammatory genes.

2. Theophylline turns on ○ HDAC2, which deacetylates histones, causing DNA to wrap tightly. 🌀

3. This represses the transcription of pro-inflammatory genes (e.g., IL-8, TNF- α).